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Learner Name: Radhika Sharma

Registration No: 12609217

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Note- The assignment Question have 5 Sections/Blocks. Kindly attempt Any **ONE question** from Each of the 5 blocks. Each question carries equal marks. For Eg:

Block 1 Attempt- Question 1a or 1b

Block 2 Attempt- Question 2a or 2b

Block 3 Attempt- Question 3a or 3b

Block 4 Attempt- Question 4a or 4b

Block 5 Attempt- Question 5a or 5b

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Q1-A: Define Symmetric Group S_4 and Show That It Is a Group Under Multiplication

Radhika Sharma

Preliminaries

Definition 1 (Permutation). Let $X = \{1, 2, 3, \dots, n\}$ be a finite set. A **permutation** of X is a bijective function $\sigma : X \rightarrow X$.

Definition 2 (Symmetric Group S_n). The **symmetric group** S_n is the set of all permutations of the set $\{1, 2, \dots, n\}$, equipped with the binary operation of **composition of functions** (also called multiplication of permutations).

$$S_n = \{ \sigma \mid \sigma : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\} \text{ is a bijection} \}$$

The group operation $\circ$ is defined by $(\sigma \circ \tau)(x) = \sigma(\tau(x))$ for all $x \in \{1, \dots, n\}$.

The Symmetric Group S_4

Definition 3 (Symmetric Group S_4). The **symmetric group** S_4 is the group of all bijections (permutations) from the set $\{1, 2, 3, 4\}$ to itself, under the operation of composition. Formally,

$$S_4 = \{ \sigma : \{1, 2, 3, 4\} \xrightarrow{\sim} \{1, 2, 3, 4\} \},$$

with $|S_4| = 4! = 24$.

Remark 1. The 24 elements of S_4 consist of:

- 1 identity permutation,
- 6 transpositions (2-cycles),
- 3 products of two disjoint transpositions,
- 8 3-cycles, and
- 6 4-cycles.

A typical element is written in *two-line notation* or in *cycle notation*. For example,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} = (1 \ 2)(3 \ 4).$$

S_4 Is a Group Under Composition

To prove that $(S_4, \circ)$ is a group, we verify the four group axioms: **Closure**, **Associativity**, **Identity**, and **Invertibility**.

Theorem 1. $(S_4, \circ)$ is a group.

Proof. **(G1) Closure.**

Let $\sigma, \tau \in S_4$. Then σ and τ are bijections from $\{1, 2, 3, 4\}$ to itself. The composition $\sigma \circ \tau$ is the composition of two bijections, which is again a bijection from $\{1, 2, 3, 4\}$ to $\{1, 2, 3, 4\}$. Hence $\sigma \circ \tau \in S_4$.

Example: Let $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2\ 4)$ in S_4 .

$$(\sigma \circ \tau)(1) = \sigma(\tau(1)) = \sigma(2) = 3, \quad (\sigma \circ \tau)(2) = \sigma(\tau(2)) = \sigma(4) = 4,$$

$$(\sigma \circ \tau)(3) = \sigma(\tau(3)) = \sigma(3) = 1, \quad (\sigma \circ \tau)(4) = \sigma(\tau(4)) = \sigma(1) = 2.$$

So $\sigma \circ \tau = (1\ 3)(2\ 4) \in S_4$. ✓

(G2) Associativity.

For all $\sigma, \tau, \mu \in S_4$ and any $x \in \{1, 2, 3, 4\}$,

$$((\sigma \circ \tau) \circ \mu)(x) = (\sigma \circ \tau)(\mu(x)) = \sigma(\tau(\mu(x))),$$

$$(\sigma \circ (\tau \circ \mu))(x) = \sigma((\tau \circ \mu)(x)) = \sigma(\tau(\mu(x))).$$

Since function composition is always associative, we have $(\sigma \circ \tau) \circ \mu = \sigma \circ (\tau \circ \mu)$ for all $\sigma, \tau, \mu \in S_4$. ✓

(G3) Existence of Identity.

Define the identity permutation $e : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$ by $e(x) = x$ for all x . Clearly e is a bijection, so $e \in S_4$. For any $\sigma \in S_4$ and all $x \in \{1, 2, 3, 4\}$,

$$(\sigma \circ e)(x) = \sigma(e(x)) = \sigma(x), \quad (e \circ \sigma)(x) = e(\sigma(x)) = \sigma(x).$$

Hence $\sigma \circ e = e \circ \sigma = \sigma$, and e is the identity element in S_4 . ✓

(G4) Existence of Inverses.

Let $\sigma \in S_4$. Since σ is a bijection, it has an inverse function $\sigma^{-1} : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$ defined by

$$\sigma^{-1}(y) = x \iff \sigma(x) = y.$$

This σ^{-1} is also a bijection, so $\sigma^{-1} \in S_4$. By definition of function inverse,

$$(\sigma \circ \sigma^{-1})(x) = \sigma(\sigma^{-1}(x)) = x = e(x),$$

$$(\sigma^{-1} \circ \sigma)(x) = \sigma^{-1}(\sigma(x)) = x = e(x),$$

so $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = e$. ✓

Example: If $\sigma = (1\ 2\ 3\ 4)$, then $\sigma^{-1} = (4\ 3\ 2\ 1) = (1\ 4\ 3\ 2) \in S_4$.

Since all four group axioms are satisfied, $(S_4, \circ)$ is a group. □

Order and Structure

The order of S_4 is $|S_4| = 4! = 24$. It is a **non-abelian** group, since composition of permutations is in general not commutative. For instance, with $\sigma = (1\ 2)$ and $\tau = (1\ 3)$:

$$\sigma \circ \tau = (1\ 3\ 2), \quad \tau \circ \sigma = (1\ 2\ 3),$$

and $(1\ 3\ 2) \neq (1\ 2\ 3)$, so $\sigma \circ \tau \neq \tau \circ \sigma$.

Hence, S_4 is a **non-abelian group of order 24** under the operation of composition of permutations. ■

Q2-B: In $G = \mathbb{Z}_{21}^* = \{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$ show that the set $H = \{1, 4, 16\}$ is a subgroup. Since G is abelian, H is automatically normal. Identify the four elements of G/H and construct their multiplication table. Is this group isomorphic to $\mathbb{Z}_4$?

Radhika Sharma

Setup

Let

$$G = \mathbb{Z}_{21}^* = \{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$$

be the multiplicative group of units modulo 21, with $|G| = \phi(21) = 12$, and let

$$H = \{1, 4, 16\}.$$

Part 1: H is a Subgroup of G

Proposition 1. $H = \{1, 4, 16\}$ is a subgroup of $G = \mathbb{Z}_{21}^*$.

Proof. We use the **Subgroup Test**: a non-empty subset H of a group G is a subgroup if and only if for all $a, b \in H$, the element $ab^{-1} \in H$.

It suffices to verify closure under multiplication and the existence of inverses within H .

Step 1: Find inverses of elements of H in G .

We compute inverses modulo 21:

$$1^{-1} \equiv 1 \pmod{21}, \quad 4^{-1} \equiv 16 \pmod{21} \text{ since } 4 \cdot 16 = 64 \equiv 1 \pmod{21},$$

$$16^{-1} \equiv 4 \pmod{21} \text{ since } 16 \cdot 4 = 64 \equiv 1 \pmod{21}.$$

All inverses lie in H . ✓

Step 2: Closure under multiplication.

We compute all products $a \cdot b \pmod{21}$ for $a, b \in H$:

$\cdot \pmod{21}$	1	4	16
1	1	4	16
4	4	16	$64 \equiv 1$
16	16	$64 \equiv 1$	$256 \equiv 4$

All products lie in H . ✓

Since H is non-empty, closed under multiplication, and closed under inverses, $H \leq G$. $\square$

Remark 1. Observe that $H = \{1, 4, 16\} = \langle 4 \rangle$, the cyclic subgroup generated by 4, since $4^1 = 4$, $4^2 = 16$, $4^3 = 64 \equiv 1 \pmod{21}$. Thus $|H| = 3$, which divides $|G| = 12$, consistent with Lagrange's Theorem.

Part 2: H is Normal in G

Proposition 2. $H \trianglelefteq G$.

Proof. Since $G = \mathbb{Z}_{21}^*$ is an **abelian group** (multiplication modulo 21 is commutative), every subgroup of G is normal. Indeed, for all $g \in G$ and $h \in H$:

$$ghg^{-1} = gg^{-1}h = h \in H.$$

Hence $gHg^{-1} = H$ for all $g \in G$, so $H \trianglelefteq G$. $\square$

Part 3: The Four Cosets of H in G

Since $|G| = 12$ and $|H| = 3$, the index $[G : H] = \frac{12}{3} = 4$, so G/H has exactly **four** cosets. We compute the left cosets $gH = \{g \cdot h \pmod{21} : h \in H\}$:

$$\begin{aligned} \mathbf{1H}: & \quad 1 \cdot \{1, 4, 16\} = \{1, 4, 16\} \\ \mathbf{2H}: & \quad 2 \cdot \{1, 4, 16\} = \{2, 8, 32 \bmod 21\} = \{2, 8, 11\} \\ \mathbf{5H}: & \quad 5 \cdot \{1, 4, 16\} = \{5, 20, 80 \bmod 21\} = \{5, 20, 17\} \\ \mathbf{10H}: & \quad 10 \cdot \{1, 4, 16\} = \{10, 40 \bmod 21, 160 \bmod 21\} = \{10, 19, 13\} \end{aligned}$$

Verification: The four cosets are pairwise disjoint and their union is:

$$\{1, 4, 16\} \cup \{2, 8, 11\} \cup \{5, 20, 17\} \cup \{10, 19, 13\} = G. \quad \checkmark$$

So the four elements of G/H are:

$$G/H = \{H, 2H, 5H, 10H\}.$$

Part 4: Multiplication Table of G/H

The group operation on cosets is defined by $(aH)(bH) = (ab)H$, computed modulo 21.

We compute each product of coset representatives and identify which coset the result falls in:

- $2 \cdot 5 = 10 \in 10H$, $2 \cdot 10 = 20 \in 5H$, $5 \cdot 10 = 50 \equiv 8 \pmod{21} \in 2H$
- $2 \cdot 2 = 4 \in H$, $5 \cdot 5 = 25 \equiv 4 \pmod{21} \in H$, $10 \cdot 10 = 100 \equiv 16 \pmod{21} \in H$

$\cdot$	H	$2H$	$5H$	$10H$
H	H	$2H$	$5H$	$10H$
$2H$	$2H$	H	$10H$	$5H$
$5H$	$5H$	$10H$	H	$2H$
$10H$	$10H$	$5H$	$2H$	H

The key observations from the table are:

- The identity element is H .
- $(2H)^2 = H$, $(5H)^2 = H$, $(10H)^2 = H$.
- Every non-identity element has order 2.

Part 5: Is $G/H \cong \mathbb{Z}_4$?

Theorem 1. $G/H \not\cong \mathbb{Z}_4$. Instead, $G/H \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (the Klein four-group V_4).

Proof. Recall that $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4. It is a **cyclic** group of order 4, containing an element of order 4.

From the multiplication table of G/H , the orders of elements are:

$$\text{ord}(H) = 1, \quad \text{ord}(2H) = 2, \quad \text{ord}(5H) = 2, \quad \text{ord}(10H) = 2.$$

Since **every non-identity element of G/H has order 2**, the group G/H contains no element of order 4. But $\mathbb{Z}_4$ has an element of order 4 (namely, the generator $1 \in \mathbb{Z}_4$).

Since $\mathbb{Z}_4$ and G/H have different element orders, they cannot be isomorphic:

$$G/H \not\cong \mathbb{Z}_4.$$

Moreover, G/H is an abelian group of order 4 in which every non-identity element has order 2. Up to isomorphism, the only such group is the **Klein four-group**:

$$V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}.$$

The isomorphism $\varphi : G/H \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$ is given explicitly by:

$$\varphi(H) = (0, 0), \quad \varphi(2H) = (1, 0), \quad \varphi(5H) = (0, 1), \quad \varphi(10H) = (1, 1).$$

One can verify that φ preserves the group operation. For example:

$$\varphi(2H \cdot 5H) = \varphi(10H) = (1, 1) = (1, 0) + (0, 1) = \varphi(2H) + \varphi(5H). \quad \checkmark$$

Hence,

$$\boxed{G/H \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \quad G/H \not\cong \mathbb{Z}_4.}$$

□

Q3-B: Find the Sylow Subgroups of D_{12}

Radhika Sharma

Preliminaries

Definition 1 (Sylow p -Subgroup). *Let G be a finite group with $|G| = p^n m$, where p is prime and $\gcd(p, m) = 1$. A **Sylow p -subgroup** of G is a subgroup of order p^n .*

Theorem 1 (Sylow's Theorem). *Let $|G| = p^n m$ with $\gcd(p, m) = 1$. Then:*

- (i) *Sylow p -subgroups exist.*
- (ii) *Any two Sylow p -subgroups are conjugate in G .*
- (iii) *The number of Sylow p -subgroups, denoted n_p , satisfies:*

$$n_p \equiv 1 \pmod{p} \quad \text{and} \quad n_p \mid m.$$

Structure of D_{12}

Definition 2 (Dihedral Group D_{12}). *The dihedral group D_{12} is the group of symmetries of a regular hexagon, consisting of 6 rotations and 6 reflections. It has order $|D_{12}| = 12$, with presentation:*

$$D_{12} = \langle r, s \mid r^6 = s^2 = e, srs^{-1} = r^{-1} \rangle.$$

The 12 elements of D_{12} are:

$$D_{12} = \{ e, r, r^2, r^3, r^4, r^5, s, sr, sr^2, sr^3, sr^4, sr^5 \}.$$

Order factorisation:

$$|D_{12}| = 12 = 2^2 \cdot 3.$$

So the relevant primes are $p = 2$ (with $p^n = 4$) and $p = 3$ (with $p^n = 3$).

Sylow 2-Subgroups of D_{12}

Here $p = 2$, $p^n = 2^2 = 4$, $m = 3$. We seek all subgroups of order 4.

Constraints on n_2 :

$$n_2 \equiv 1 \pmod{2} \quad \text{and} \quad n_2 \mid 3.$$

So $n_2 \in \{1, 3\}$.

Finding the Sylow 2-subgroups. We look for subgroups of order 4 in D_{12} . The subgroup of rotations $\langle r \rangle = \{e, r, r^2, r^3, r^4, r^5\} \cong \mathbb{Z}_6$ has a unique subgroup of order 2, namely $\{e, r^3\}$. We combine this with reflections to form groups of order 4.

The three Sylow 2-subgroups of D_{12} are:

$$P_1 = \{e, r^3, s, sr^3\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

$$P_2 = \{e, r^3, sr, sr^4\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

$$P_3 = \{e, r^3, sr^2, sr^5\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Proposition 1. P_1, P_2, P_3 are subgroups of D_{12} , each of order 4.

Proof. We verify $P_1 = \{e, r^3, s, sr^3\}$. Using the relations $r^6 = e, s^2 = e, srs^{-1} = r^{-1}$:

- $(r^3)^2 = r^6 = e$. ✓
- $s^2 = e$. ✓
- $(sr^3)^2 = sr^3 \cdot sr^3 = s(r^3s)r^3 = s \cdot sr^{-3} \cdot r^3 = s^2 \cdot e = e$. ✓
- $r^3 \cdot s = sr^{-3} = sr^3$ (since $r^{-3} = r^3$ in $\langle r \rangle \cong \mathbb{Z}_6$). So $r^3 \cdot s = sr^3 \in P_1$. ✓

Closure holds for all products; all elements are self-inverse. Hence $P_1 \leq D_{12}$. The verifications for P_2 and P_3 are analogous, replacing s with sr and sr^2 respectively. $\square$

Remark 1. Since $n_2 = 3 \neq 1$, **no Sylow 2-subgroup is normal** in D_{12} . The three subgroups P_1, P_2, P_3 are mutually conjugate.

Sylow 3-Subgroups of D_{12}

Here $p = 3, p^n = 3^1 = 3, m = 4$. We seek all subgroups of order 3.

Constraints on n_3 :

$$n_3 \equiv 1 \pmod{3} \quad \text{and} \quad n_3 \mid 4.$$

So $n_3 \in \{1, 4\}$. But $n_3 \mid 4$ and $n_3 \equiv 1 \pmod{3}$, giving $n_3 = 1$ (since $4 \not\equiv 1 \pmod{3}$).

Hence there is exactly **one** Sylow 3-subgroup, which must be normal in D_{12} .

Finding the Sylow 3-subgroup. Consider the rotation subgroup $\langle r \rangle \cong \mathbb{Z}_6$. It contains the unique subgroup of order 3:

$$Q = \langle r^2 \rangle = \{e, r^2, r^4\} \cong \mathbb{Z}_3.$$

Proposition 2. $Q = \{e, r^2, r^4\}$ is a normal subgroup of D_{12} .

Proof. Q is a subgroup since $r^2 \cdot r^2 = r^4, r^2 \cdot r^4 = r^6 = e$, and $r^4 \cdot r^4 = r^8 = r^2$; it is clearly closed and every element is invertible within Q . Since $n_3 = 1$, Q is the unique Sylow 3-subgroup and hence is normal: $gQg^{-1} = Q$ for all $g \in D_{12}$.

We verify normality directly for the generator s :

$$sr^2s^{-1} = r^{-2} = r^4 \in Q, \quad sr^4s^{-1} = r^{-4} = r^2 \in Q.$$

Since Q is also closed under conjugation by r (as r commutes with powers of r), we have $gQg^{-1} = Q$ for all $g \in D_{12}$. Hence $Q \trianglelefteq D_{12}$. $\square$

Summary

Prime p	$ P $	n_p	Sylow p -Subgroup(s)	Normal?
			$P_1 = \{e, r^3, s, sr^3\}$	
2	4	3	$P_2 = \{e, r^3, sr, sr^4\}$	No
			$P_3 = \{e, r^3, sr^2, sr^5\}$	
3	3	1	$Q = \{e, r^2, r^4\}$	Yes

To summarise:

- D_{12} has **three** Sylow 2-subgroups of order 4, each isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. They are not normal.
- D_{12} has a **unique** Sylow 3-subgroup of order 3, namely $Q = \langle r^2 \rangle \cong \mathbb{Z}_3$, which is normal in D_{12} .

Q4-B: Find All Ideals in the Ring $\mathbb{Z}_{18}$. Generalize to $\mathbb{Z}_n$

Radhika Sharma

Preliminaries

Definition 1 (Ring and Ideal). A **ring** $(R, +, \cdot)$ is a set with two binary operations satisfying the ring axioms. A non-empty subset $I \subseteq R$ is an **ideal** of R if:

- (i) $(I, +)$ is a subgroup of $(R, +)$, and
- (ii) For all $r \in R$ and $a \in I$: $ra \in I$ and $ar \in I$ (absorption property).

Definition 2 (Principal Ideal). For a commutative ring R with unity and any element $a \in R$, the **principal ideal generated by a** is

$$\langle a \rangle = aR = \{ ar \mid r \in R \}.$$

Theorem 1. $\mathbb{Z}_n$ is a commutative ring with unity. Every ideal of $\mathbb{Z}_n$ is a principal ideal. Specifically, the ideals of $\mathbb{Z}_n$ are exactly the subgroups of $(\mathbb{Z}_n, +)$, which are in bijection with the divisors of n .

Proof sketch. Since $\mathbb{Z}_n$ is a principal ideal ring (a quotient of the PID $\mathbb{Z}$), every ideal of $\mathbb{Z}_n$ is principal. Moreover, every additive subgroup of $\mathbb{Z}_n$ is automatically an ideal, since for any $a \in I \subseteq \mathbb{Z}_n$ and $r \in \mathbb{Z}_n$, the product ra is a repeated sum of a , hence lies in I . The additive subgroups of $\mathbb{Z}_n$ correspond bijectively to divisors d of n via $d \mapsto \langle d \rangle$. $\square$

Part 1: All Ideals of $\mathbb{Z}_{18}$

The ring $\mathbb{Z}_{18} = \{0, 1, 2, \dots, 17\}$ has order 18. We first find all divisors of 18:

$$18 = 2 \cdot 3^2, \quad \text{divisors of 18: } 1, 2, 3, 6, 9, 18.$$

Each divisor d of 18 gives a unique ideal $\langle d \rangle$ of $\mathbb{Z}_{18}$.

For each divisor $d \mid 18$, the ideal $\langle d \rangle$ consists of all multiples of d modulo 18, and has order $18/d$:

1. $d = 1$:

$$\langle 1 \rangle = \mathbb{Z}_{18} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17\}, \quad |\langle 1 \rangle| = 18.$$

This is the **improper (whole ring) ideal**.

2. $d = 2$:

$$\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14, 16\}, \quad |\langle 2 \rangle| = 9.$$

This is the ideal of all even residues modulo 18.

3. $d = 3$:

$$\langle 3 \rangle = \{0, 3, 6, 9, 12, 15\}, \quad |\langle 3 \rangle| = 6.$$

4. $d = 6$:

$$\langle 6 \rangle = \{0, 6, 12\}, \quad |\langle 6 \rangle| = 3.$$

5. $d = 9$:

$$\langle 9 \rangle = \{0, 9\}, \quad |\langle 9 \rangle| = 2.$$

6. $d = 18$:

$$\langle 18 \rangle = \langle 0 \rangle = \{0\}, \quad |\langle 18 \rangle| = 1.$$

This is the **zero ideal** (trivial ideal).

Hence $\mathbb{Z}_{18}$ has exactly **six ideals**:

Ideal	Generator d	Elements	Order
$\langle 1 \rangle$	1	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17\}$	18
$\langle 2 \rangle$	2	$\{0, 2, 4, 6, 8, 10, 12, 14, 16\}$	9
$\langle 3 \rangle$	3	$\{0, 3, 6, 9, 12, 15\}$	6
$\langle 6 \rangle$	6	$\{0, 6, 12\}$	3
$\langle 9 \rangle$	9	$\{0, 9\}$	2
$\langle 0 \rangle$	0	$\{0\}$	1

Remark 1. There is a natural **inclusion-reversing** (containment) relationship among the ideals, reflecting the divisibility lattice of 18:

$$\langle 0 \rangle \subset \langle 9 \rangle \subset \langle 3 \rangle \subset \langle 1 \rangle = \mathbb{Z}_{18},$$

$$\langle 0 \rangle \subset \langle 6 \rangle \subset \langle 2 \rangle \subset \langle 1 \rangle = \mathbb{Z}_{18},$$

$$\langle 0 \rangle \subset \langle 6 \rangle \subset \langle 3 \rangle \subset \langle 1 \rangle = \mathbb{Z}_{18}.$$

In general, $\langle d_1 \rangle \subseteq \langle d_2 \rangle$ if and only if $d_2 \mid d_1$.

Part 2: Generalization to $\mathbb{Z}_n$

Theorem 2 (Ideals of $\mathbb{Z}_n$). *The ideals of $\mathbb{Z}_n$ are in bijection with the positive divisors of n . Explicitly:*

(i) For every divisor $d \mid n$, the set

$$\langle d \rangle = \{0, d, 2d, 3d, \dots, n - d\}$$

is an ideal of $\mathbb{Z}_n$, with $|\langle d \rangle| = n/d$.

(ii) Every ideal of $\mathbb{Z}_n$ is of this form.

(iii) There are exactly $\tau(n)$ ideals in $\mathbb{Z}_n$, where $\tau(n)$ denotes the number of positive divisors of n .

Proof. (i) $\langle d \rangle$ is an ideal.

Let $d \mid n$. The set $\langle d \rangle = d\mathbb{Z}_n = \{0, d, 2d, \dots, (n/d - 1)d\}$ is a cyclic subgroup of $(\mathbb{Z}_n, +)$ of order n/d . To verify the absorption property: for any $r \in \mathbb{Z}_n$ and $a = kd \in \langle d \rangle$,

$$r \cdot a = r \cdot kd = (rk)d \equiv (rk \bmod n/d) \cdot d \pmod{n},$$

which is again a multiple of d , hence in $\langle d \rangle$. Thus $\langle d \rangle$ is an ideal.

(ii) Every ideal of $\mathbb{Z}_n$ has this form.

Let I be any ideal of $\mathbb{Z}_n$. Then I is in particular an additive subgroup of $\mathbb{Z}_n$. By the classification of subgroups of cyclic groups, every subgroup of $\mathbb{Z}_n$ is cyclic, generated by some divisor d of n :

$$I = \langle d \rangle \quad \text{for some } d \mid n.$$

Since every additive subgroup of $\mathbb{Z}_n$ is automatically an ideal (as $\mathbb{Z}_n$ is commutative and multiplication by any r sends multiples of d to multiples of d), the ideals and additive subgroups of $\mathbb{Z}_n$ coincide.

(iii) Count.

The divisors of n are in bijection with the ideals of $\mathbb{Z}_n$, so the number of ideals equals $\tau(n)$, the number of positive divisors of n . $\square$

Corollary 1 (Inclusion of Ideals). For divisors d_1, d_2 of n :

$$\langle d_1 \rangle \subseteq \langle d_2 \rangle \iff d_2 \mid d_1.$$

In particular:

- The largest ideal is $\langle 1 \rangle = \mathbb{Z}_n$ (the whole ring).
- The smallest ideal is $\langle n \rangle = \langle 0 \rangle = \{0\}$ (the zero ideal).

Corollary 2 (Maximal and Prime Ideals of $\mathbb{Z}_n$). An ideal $\langle d \rangle$ of $\mathbb{Z}_n$ is:

- **Maximal** if and only if $d = p$ is prime and $p \mid n$. In this case $\mathbb{Z}_n/\langle p \rangle \cong \mathbb{Z}_p$ is a field.
- **Prime** if and only if $d = p$ is prime and $p \mid n$.

(In $\mathbb{Z}_n$, the maximal ideals and prime ideals coincide.)

Example 1 (Verification for $\mathbb{Z}_{18}$). We have $n = 18 = 2 \cdot 3^2$, so $\tau(18) = (1+1)(2+1) = 6$, confirming the six ideals found in Part 1. The maximal (and prime) ideals are $\langle 2 \rangle$ and $\langle 3 \rangle$, corresponding to the prime divisors 2 and 3 of 18.

Theorem 3 (General Formula for Ideals of $\mathbb{Z}_n$). If $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ is the prime factorisation of n , then the number of ideals of $\mathbb{Z}_n$ is

$$\tau(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

Each ideal is of the form $\langle d \rangle$ for a unique divisor d of n , and consists of all multiples of d modulo n :

$$\langle d \rangle = \{0, d, 2d, \dots, (n/d - 1) \cdot d\}, \quad |\langle d \rangle| = \frac{n}{d}.$$

Summary

Ring	Number of Ideals	Ideals
$\mathbb{Z}_{18}$	$\tau(18) = 6$	$\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 6 \rangle, \langle 9 \rangle, \langle 0 \rangle$
$\mathbb{Z}_n$, general	$\tau(n)$	$\langle d \rangle$ for each $d \mid n$
$\mathbb{Z}_p$, p prime	2	$\langle 0 \rangle$ and $\langle 1 \rangle = \mathbb{Z}_p$ only

The last row reflects the well-known fact that $\mathbb{Z}_p$ is a **field** (hence a simple ring) whenever p is prime, having no proper non-trivial ideals. ■

Q5-A: Prove That a Group of Order 40 Has a Nontrivial Normal Subgroup

Radhika Sharma

Preliminaries

Definition 1 (Normal Subgroup). A subgroup $N \leq G$ is **normal** in G , written $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$. It is **nontrivial** if $N \neq \{e\}$ and $N \neq G$.

Theorem 1 (Sylow's Theorem). Let G be a finite group with $|G| = p^n m$, where p is prime and $\gcd(p, m) = 1$. Let n_p denote the number of Sylow p -subgroups of G . Then:

- (i) Sylow p -subgroups exist and have order p^n .
- (ii) Any two Sylow p -subgroups are conjugate in G .
- (iii) $n_p \equiv 1 \pmod{p}$ and $n_p \mid m$.
- (iv) A Sylow p -subgroup is normal in G if and only if $n_p = 1$.

Proof

Theorem 2. Every group G of order 40 has a nontrivial normal subgroup.

Proof. **Step 1: Factor the order and apply Sylow's Theorem.**

We have

$$|G| = 40 = 2^3 \cdot 5.$$

Apply Sylow's Theorem with $p = 5$, so that $p^n = 5$ and $m = 8$. The number of Sylow 5-subgroups satisfies:

$$n_5 \equiv 1 \pmod{5} \quad \text{and} \quad n_5 \mid 8.$$

Step 2: Determine n_5 .

The positive divisors of 8 are 1, 2, 4, 8. We check each modulo 5:

$$1 \equiv 1, \quad 2 \equiv 2, \quad 4 \equiv 4, \quad 8 \equiv 3 \pmod{5}.$$

The *only* divisor of 8 satisfying $n_5 \equiv 1 \pmod{5}$ is $n_5 = 1$. Hence:

$$n_5 = 1.$$

Step 3: Conclude normality.

Since $n_5 = 1$, there is a **unique** Sylow 5-subgroup $P_5 \leq G$. By Sylow's Theorem (iv), uniqueness implies normality:

$$P_5 \trianglelefteq G.$$

Since $|P_5| = 5$, we have $P_5 \neq \{e\}$ and $P_5 \neq G$, so P_5 is a nontrivial normal subgroup of G . $\square$

Conclusion

Every group of order 40 has a nontrivial normal subgroup.

Corollary 1. *No group of order 40 is simple.*

Proof. A simple group has no proper nontrivial normal subgroups. Since every group of order 40 contains $P_5 \trianglelefteq G$ with $|P_5| = 5$, no group of order 40 can be simple. $\square$

Remark 1. The argument is entirely elementary: among the divisors 1, 2, 4, 8 of 8, exactly one — namely 1 — is congruent to 1 modulo 5. This forces $n_5 = 1$ without any further case analysis, and the unique Sylow 5-subgroup is immediately normal.